

The Ancestry Recording Monitor: Definitions, the 2026-07-30 Experiment Battery, and What It Established

local_mixing working notes

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Abstract

The ancestry monitor is `fmix`'s instrument for *material transport*: it tracks, for every gate of the evolving circuit, the set of original input gates its material descends from. This note documents the monitor itself — the recording rules and the definition and rationale of every reported measure — and then gives a complete account of the experiment battery run against it on 2026-07-29/30: the A/B/C $\times$ twist-rate campaign, the bracket-placer factorial, the recording-semantics isolation, the conditional census that audited the instrument, the functional cross-checks that bounded its meaning, and the slice-4 sanity battery. Conclusions: the monitor is sound (its headline number was already immune to the suspected dilution artifact); its readings are genuinely about transport, not about feasible-adversary legibility — a $100\times$ spread in `anc` across twist arms has no counterpart in any functional channel measured; and its cross-run comparisons are valid at matched effective work within one input universe, with saturation the main caveat across universes.

1 The monitor

1.1 Recording layer

Litters. Every gate carries a *litter* tag: the replacement event that created it. Input gates and born-random material are singleton litters; splits and merges *propagate* the parent's id (a litter fragments under churn rather than being reassigned); a DB splice stamps all its products with one fresh id.

Ancestor sets. Per litter, a bitset over the m input-gate indices: the union of the input gates whose material fed that litter. Recording rules:

- input gate i starts as the implicit singleton $\{i\}$ (litter ids $< m$ read as singletons unless an explicit entry overrides; fresh ids start at m , so no aliasing);
- a DB splice's product litter gets the *union* of the consumed window's litters' sets; catalogue merges likewise union their two parents;
- born-random material (ORIGIN_SYNT: copy-pair inserts, twist brackets) starts *empty* — it carries no input material;
- since placer v2 (commit 83ae0d11), a g57 twist seam that consumes context gates unions the consumed litters into the bracket word's litter, exactly like a DB splice. (v1 dropped them; the isolation experiment in §2.4 measures what that was worth.)

The union can only grow: unlike the scalar `origin` label — which a mixed-lineage splice destroys (the `osyn` fraction measures exactly that erosion) — an ancestor set never loses information. That is the monitor’s reason to exist: it keeps answering “what is this gate made of, and how far has input material travelled” long after per-gate origin labels have become meaningless.

Cost envelope. Exact mode (`-ancestors`) stores m bits per litter and refuses to arm above 20,000 input gates. Sampled mode (`-anc-samples K`, from the parallel monitoring session, commit `8b73a98c`) tracks only K randomly chosen input gates (“tracers”), K bits per litter, which removes the size ceiling; the tracer set is a pure function of $(m, K, \text{anc_sample_seed})$ and survives resumes. Calibration on an identical trajectory: the sampled estimate of `anc` agrees with exact within 3% at $K=256$ and 8% at $K=64$.

1.2 The measures

anc — mean ancestor-set cardinality over live gates, *conditional on a non-empty set* (gates whose set is empty are skipped — established by reading `anc_stats` and confirmed by the census of §2.5). Rationale: “what is a mixed gate made of” — the width of the input window a gate’s material has been assembled from.

ancspan — mean over the same gates of $(\max - \min)/(m - 1)$ of the set’s input indices: how far apart in the *original* circuit the meeting material originated. The report line carries the normalised form; the ancestry report prints the raw mean (“span (input gates)”) plus log-bucketed histograms of per-gate cardinality, span, and per-input fanout. Rationale: **anc** can grow by merging neighbours; **span** only grows when *distant* input material actually meets — it is the transport-distance reading.

fanout/input — per original gate, how many live gates descend from it (the transpose view; its histogram exposes extinct inputs vs. super-spreaders).

Sampled-mode readouts (tracer_report): **desc** = exact per-tracer fanout (unbiased for mean fanout since each input has inclusion probability K/m); from it **est anc** = **desc** · m/size recovers the exact-mode **anc** without storing m bits anywhere. **cov** = fraction of 64 equal position-buckets of the *current* circuit containing at least one descendant of a tracer; **ent** = normalised entropy of the descendant positions over those buckets. **cov/ent** are the security-facing pair — they ask directly whether an adversary can *localise* where an original gate went — and they are natively samplable, unlike **span**, which a sample can only underestimate. In sampled mode the **anc=/ancspan=** fields are deliberately left at zero rather than silently changing meaning.

2 The experiment battery

Common setup unless noted: the 20,000-gate, 64-wire sample (`g16_20k.mpmct1`, the first $\approx 3/7$ of a Gray-fold gadgetized sliced sandwich, $n=16$), seed 101, 2M moves, C schedule (`-p-mix 0.1`), exact ancestry on, server directory `~/tds/menucal_20260728`. “Effective work” (`eff`) is the trapezoid integral of moves per gate — the dose axis that removes size-inflation bias. Legacy = shipped 3-CNOT swap-family brackets; `g57 v1` = anchor-first all-`g57` brackets; `v2` = `v1` + edge slide + joint acceptance + ancestry union.

2.1 E1: A/B/C × twist rate, legacy vs. `g57 v1`

Fifteen runs: three schedules (A phased, B 20% MIX, C 10% MIX) × {no twist, legacy, `g57 v1`} × rates {0.002, 0.01}. C-arm finals:

arm	size	anc	span	polf
no twist	48,493	7,941	8,626	0.000
legacy 0.002	133,665	425	1,111	0.422
legacy 0.01	262,115	62	369	0.487
g57v1 0.002	86,696	2,604	3,374	0.000
g57v1 0.01	216,281	637	1,341	0.000

At matched eff ≈ 32 : no-twist **anc** 3,961, g57 2,170 (-45%), legacy 425 (-89%). **Result:** both twist kinds suppress transport at equal dose; the g57 brackets suppress $5\text{--}6\times$ less while also ending smaller. A and B schedules reproduce the ordering.

2.2 E2: placer factorial

C arm, four placer configurations via the env kill-switches (v1 / slide-only / retry-only / both), both rates. At 0.01: mean net gates per seam $4.51 \rightarrow 4.38 \rightarrow 3.93 \rightarrow 3.72$; size $216\text{k} \rightarrow 168\text{k}$; **anc** $637 \rightarrow 921$ (v1 $\rightarrow$ both). **Result:** joint acceptance (window redraw unless both seams land) is the larger transport lever; the slide composes.

2.3 E3: v2 full trio

Six runs, all schedules $\times$ both rates, full v2. C-arm finals: size $78,233$ / **anc** $3,038$ / span $3,895$ at 0.002; $168,192$ / 921 / $1,625$ at 0.01. **Result:** v2 vs. legacy at 0.002 = $7\times$ the transport in a 41% smaller circuit.

2.4 E4: recording-semantics isolation

Six runs with v2's placement disabled (both env switches) so only the *ancestry-union* recording change differs from E1's g57 v1 arms. Sizes match v1 within 1% (clean isolation); **anc** moves by only $+1.5\text{--}10\%$ (C 0.002: $2,604 \rightarrow 2,716$; C 0.01: $637 \rightarrow 698$). **Result:** of the v1 $\rightarrow$ v2 **anc** gain, $\sim 10\%$ is the fairer recording, $\sim 90\%$ is placement. The fix matters more as consumption rates rise (v1 consumed only ~ 0.8 gates/twist).

2.5 E5: the conditional census (instrument audit)

Conjecture under test: the **anc** drop under twists is an artifact — bracket gates carry no ancestry, so means are deflated. A state-file census (litters + ancestor sets parsed directly) recomputed, for all fifteen E1 states: the unconditional mean, the conditional mean over non-empty sets, and the empty-set fraction. Extract:

state	anc _{all}	anc _{cond}	f_{empty}	f_{synth}
expC (no twist)	7941.4	7941.4	0.000	0.993
C_tw002 (legacy)	378.6	424.6	0.108	0.998
A_tw01 (legacy)	37.8	69.4	0.455	0.997
C_g57tw002	2527.4	2604.1	0.029	0.996
C_g57tw01	505.6	636.6	0.206	0.998

Result: the reported **anc** *already equals the conditional mean* (the reporter skips empty sets), so the simple dilution form of the conjecture is refuted; conditioning changes no ordering. The empty-lineage *population* is nonetheless real at high rates (up to 46% legacy at 0.01) — brackets do dilute the material, just not the reported statistic. ($f_{\text{synth}} \approx 0.99$ everywhere is the old **osyn** story — scalar origins die under ordinary churn — and is what the set-based monitor exists to survive.)

2.6 E6: functional cross-checks (what anc does not measure)

Affine-reconstruction plates of the input’s prefix progress against each mixed circuit, plus the stress battery, on the seven C-family arms: degree-1 plates are identical across all arms ($\rho = 1.000$, $z \approx 7.1$) — and are provably *blind* to twists (every twist type is an affine conjugation; the measure is affine-invariant). Degree-2 plates (not blind): at ceiling in every arm ($\rho = 1.000$, $z \approx 4.5$), depth differences within noise. Battery statistical probes: prominence 0.26–0.31 in every arm, progress monotonicity $z = 7$ –9 everywhere. **Result:** a $100\times$ spread in **anc** across arms has *no counterpart* in any measured feasible-adversary channel. **anc** is a transport gauge, not a legibility gauge; twist-arm **anc** gaps must not be read as security deltas. (The one twist-sensitive channel found — the syntactic write-census gap — reversed direction on slice 4, E7, so it is not robust either.)

2.7 E7: slice-4 sanity battery

The same seven arms on the *middle seventh* of the source gadget (6,640 gates containing the gadgetized N-band / C|D seam), at matched moves per gate. Transport results (finals; anc as fraction of the input universe in parentheses):

arm	size ($\times$ input)	anc	span frac
no twist	2.23	6,464 (0.97)	0.99
legacy 0.002	6.66	697 (0.105)	0.229
legacy 0.01	13.1	81 (0.012)	0.064
g57v1 0.002	4.13	2,927 (0.44)	0.528
g57v2 0.002	3.59	3,669 (0.55)	0.642
g57v2 0.01	7.95	1,140 (0.18)	0.253

Result: size-growth ratios are slice-independent to a few percent; the arm ordering replicates exactly; but all *anc fractions* run ~ 1.4 – $2\times$ higher than on the prefix sample because the 6,640-gate ancestor universe saturates (the no-twist arm reaches 97% of it, span 99%). Legibility instruments, by contrast, do see the slice difference: every arm is more affine-recoverable (deg-1 mean H 0.31–0.34 vs. 0.38–0.41; battery parity prominence doubled) — the N-band’s linear material survives as a softer region regardless of twist treatment.

3 Conclusions

1. **The monitor is sound.** Its headline statistic was already conditional (immune to the empty-set dilution suspected of it), and its one genuine recording unfairness — twist seams dropping consumed lineage — is fixed and measured at only +1.5–10%. Readings replicate across schedules and slices.
2. **What it measures is transport, and only transport.** Twists genuinely suppress input-material transport per unit work (–45% g57, –89% legacy at matched eff) — that is a true statement about material flow. But no functional instrument (degree-1/2 affine, statistical probe battery) registers any corresponding legibility change. Until an adversary class is exhibited that tracks **anc**, twist-arm **anc** gaps are economics, not security.
3. **Reading rules.** Compare **anc**/**span** only at matched effective work; within one input universe, absolute values are fine; across universes, use fractions and beware saturation (slice 4); remember both are conditional means and check the empty fraction (f_{empty} , or **carriers** in

sampled mode) when twist or insert rates are high; for localisation questions use the sampled `cov/ent`, which have no exact-mode analogue.

4. **Instrument portfolio.** The battery only worked because the monitor was cross-examined: the state-file census audited the reporter, the affine plates exposed their own blindness (twists are affine), and the degree-2/battery probes bounded the functional meaning. That cross-examination pattern — instrument, audit, ground-truth — is the reusable method here.
5. **Production.** Exact mode stays a small-input instrument; `-anc-samples 256` is the production-scale monitor (3% calibration), with `est anc` recovering this note’s headline quantity and `cov/ent` adding the localisation reading.

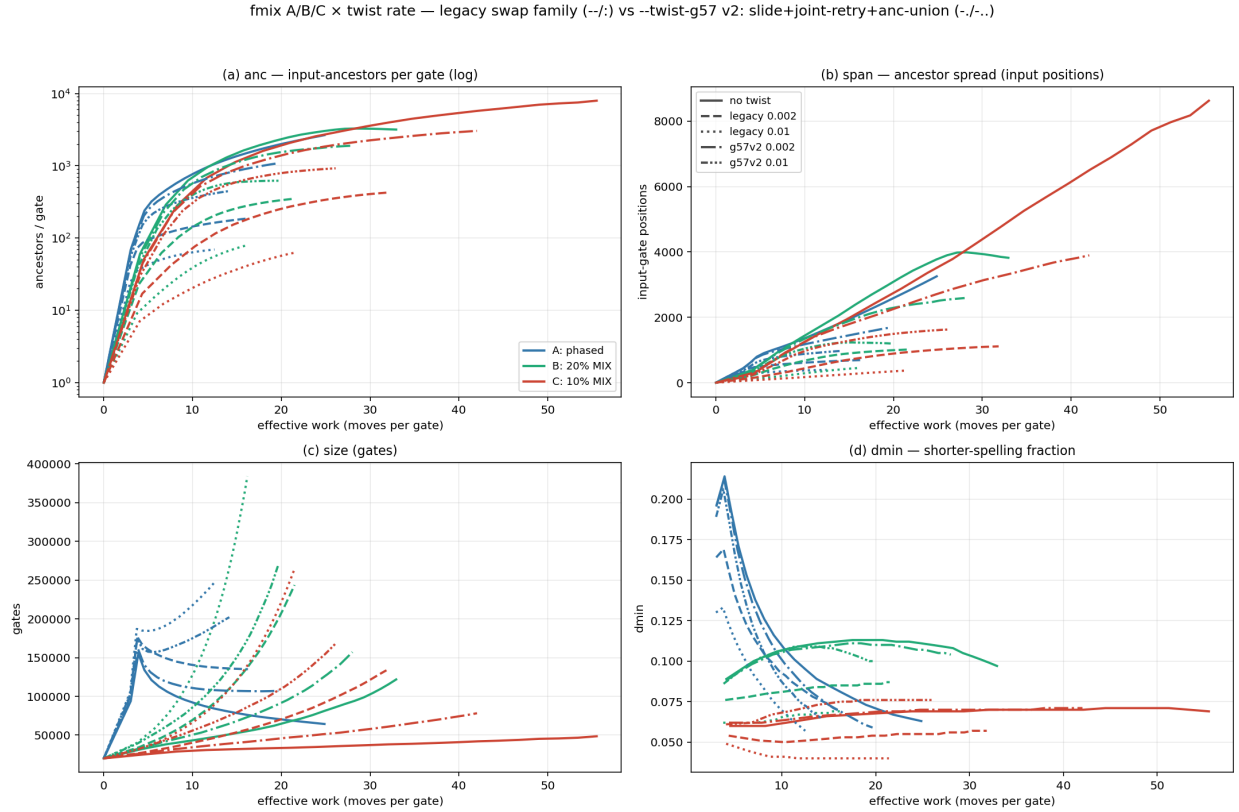


Figure 1: E1/E3 combined: anc, span, size, dmin vs. effective work; A/B/C colours, twist variant by line style.